

8 Bears are several species of mammal in the family Ursidae.

Climate change and loss of habitats due to human activity have reduced the size of some bear populations in recent years.

- (a) The International Union for Conservation of Nature (IUCN) has a role in the conservation of species, such as species of bear.

Outline the role of the IUCN in the conservation of species.

.....

.....

.....

.....

.....

.....

..... [3]

- (b) Grizzly bears, *Ursus arctos horribilis*, and polar bears, *Ursus maritimus*, are found in North America.

Fig. 8.1 shows a grizzly bear and a polar bear.



grizzly bear



polar bear

Fig. 8.1

Grizzly bears have:

- a varied diet that includes grasses, roots, berries, nuts, rodents, insects and fish
- a habitat of woodland or grassland
- a maximum mass of 270 kg.

Polar bears have:

- a diet that consists of other animals, such as seals
- a habitat of land and sea ice
- a maximum mass of 450 kg.

- (c) (i) Human activity has disturbed the habitat of many populations of spectacled bear, *Tremarctos ornatus*. The spectacled bear feeds on a variety of plant and animal species.

Scientists compared the diversity of the food eaten by two populations of spectacled bear by analysing their faeces. One population lived in an undisturbed habitat and the other population lived in a habitat that had been disturbed by road building.

The scientists analysed the faeces from 60 bears in each population. They identified the species present and the number of individuals of each species in the faeces.

The scientists calculated the diversity of species in the faeces of the two populations using Simpson's Index of Diversity.

The formula for Simpson's Index of Diversity is:

$$D = 1 - \left(\sum \left(\frac{n}{N} \right)^2 \right)$$

n = number of individuals of each species present in the sample

N = total number of all individuals of all species

Table 8.1 shows the results and some of the steps for calculating the D value for the disturbed population.

Table 8.1

species eaten by bear	number of individuals	$\frac{n}{N}$	$\left(\frac{n}{N}\right)^2$
<i>Gracilinanus aceramarcae</i>	3	0.048	0.002
<i>Oligorizomys</i> sp.	5	0.081	0.007
<i>Hesperomeles cuneata</i>	4	0.065	0.004
<i>Puya atra</i>	28	0.452	0.204
<i>Vaccinium floribundum</i>	2	0.032	0.001
<i>Pernettya prostrata</i>	8	0.129	0.017
<i>Gaultheria hapalotricha</i>	12		
total	62		

Calculate Simpson's Index of Diversity by completing Table 8.1 in the spaces provided.

Write the value for Simpson's Index of Diversity on the dotted line.

Simpson's Index of Diversity (D) = [3]

- (ii) The scientists calculated a D value of 0.833 for the undisturbed population.

Explain what is shown by the difference in the D values between the two populations of bear.

.....

..... [1]

[Total: 13]